

Consequences for the Information Question: Magnitude, Phase and the Second Channel

The photon as the exact pure-phase/connection anchor, Vopson/Landauer as the
magnitude channel, and what only the geometry holds

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30 June 2026 | Doc. 290

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- α coupling of the bare E_0 : $\alpha = \xi E_0^2 \rightarrow \xi E_0^2 / K_{\text{frak}}$; +0.68 % is $K_{\text{frak}}^{-1/2}$.
- Fractal sub-steps: " $\approx 13 \sqrt{2}$ steps per factor 7500 = I_1 " annotated – $(\sqrt{2})^{13} = 90.5$; required are $25.7 \sqrt{2}$ steps or 12.87 factor-2 steps.

Abstract

The magnitude/phase decomposition of the FFGFT mass operator (Doc. 288/289) reorganises the information question. Information has a *magnitude side* (mass/energy – Landauer and Vopson, derived by FFGFT from the T^4 geometry) and an independent *phase side* (holonomy, mixing – the one open datum θ). The cleanest witness is the **photon**: exactly massless – zero rest-mass magnitude, hence pure phase in the mass channel – and the quantum of the $U(1)$ gauge potential A_μ , which is the connection; FFGFT's information unit $E_{\text{bit}} = \hbar c / L$ is a photon energy. The *amount* of information is not representation-relative but absolute and dimensionless, anchored in ξ : $I_1 = \log_2(1/\xi) \approx 12.87$ bit per degree of freedom; energy and capacity are its dimensional, frequency-scaling dressings. The question is not "is information fundamental?" but **which channel** – and only the geometry holds both. A structural diagnosis, not a new derivation.

.1 Starting point: geometry first, and the magnitude/phase lens

FFGFT's stance on the information question is documented and geometry-first. Doc. 243 (discussion with Onur Teker, UIFT): the map of FFGFT into an information formalism is a *projection* π , not an isomorphism; the bit count $I = \log_2 |\mathcal{W}|$ presupposes the geometric state space. Doc. 255/257 derive Landauer's principle and Vopson's mass-information equivalence from the T^4 geometry: a bit is a minimal topological winding $\Delta w = 1$, "information is topologically frozen kinetic energy", $E_{\text{bit}}(L) = \hbar c / L = pc$. Doc. 274 places FFGFT *geometry-primitive* beside information-primitive (Vopson). FFGFT does not reject Vopson – it grounds him.

The new lens is the magnitude/phase decomposition of the mass operator (Doc. 288/289): the operator is a Z_3 -circulant, DFT-diagonalised, and its content splits into a *magnitude spectrum* (masses, energy) and a *phase spectrum* (mixing, holonomy). The θ finding (Doc. 282) says the magnitude spectrum fixes the phase spectrum only up to an all-pass factor (made precise in §2). That is the key to all that follows.

.2 Two channels: magnitude and phase

$\Delta w = 1$ and $E_{\text{bit}} = \hbar c / L$ are magnitude/energy statements. Since the magnitude spectrum fixes the phase spectrum only up to an all-pass factor (see below):

Vopson's mass-information equivalence – which FFGFT derives – structurally captures *only the magnitude channel*. The phase channel (holonomy, mixing, θ) is independent information, not reducible to mass/energy.

This shows concretely in two places. *First, mixing-blindness*: a mass operator $M = U \text{diag}(m) U^\dagger$ carries the eigenvalues m (magnitude channel) and the eigenvectors U (mixing, phase channel); any function of the eigenvalues alone ($\sum m$, Koide Q) is invariant under U .

Numerically (`masse_information_phasenblind.py`): for fixed masses (0.5, 1.5, 3.0), $\sum m = 5$ and $Q = 0.372$ stay equal across three mixings while $\|M_{\text{offdiag}}\|$ (1.42 to 1.78) varies – two physically distinct operators with the same mass-information.

Second, the phase carries the structure: known in signal engineering since Oppenheim/Lim (1981) – swapping the magnitude and phase spectra of two patterns, the reconstruction follows the *phase* donor. Numerically (`phase_traegt_struktur.py`): phase-only correlates with the original (+0.53), magnitude-only does not (−0.24). A mass-energy accounting thus captures the *structurally poorer* channel.

Third, precisely: “the magnitude does not fix the phase” holds only *up to an all-pass factor*. Every stable transfer function factors as $H = H_{\text{min}} \cdot H_{\text{ap}}$: for the *minimum-phase* part the magnitude does fix the phase (Bode/Hilbert relation, $\arg = \mathcal{H}\{\log |H|\}$), whereas the *all-pass* ($|H_{\text{ap}}| = 1$) carries a phase the magnitude does *not* fix. Numerically (`minimalphase_allpass.py`): the phase reconstructed from the magnitude alone matches $\arg H_{\text{min}}$ (error $\sim 10^{-15}$), the residual is exactly the all-pass. Free – and thus the open part – is the *all-pass*, and that is where the θ holonomy sits (the minimum-phase (FFGFT) $\times$ all-pass (HLV-*R*) reading, Doc. 288).

.3 The photon: the exact anchor

The cleanest witness is the photon – and the corpus already holds the pieces. Doc. 007 treats neutrinos via the photon analogy: photon $E^2 = (pc)^2 + 0$ (exactly massless), neutrino $E^2 = (pc)^2 + (\sqrt{\xi^2/2} mc^2)^2$ (quasi-massless). The photon is the $m = 0$ limit, the neutrino “photon plus the first $\xi^2/2$ correction”.

The photon has rest-mass magnitude exactly zero – the exact all-pass *in the mass channel*: all of its rest-mass content is phase. (Its energy $\hbar\omega$ remains a magnitude quantity; “pure phase” refers to the rest-mass, not the energy channel.) Three things coincide:

1. **Unfrozen information.** The photon is the unfrozen case of “information = frozen kinetic energy” ($E = pc$); FFGFT’s unit $E_{\text{bit}} = \hbar c/L$ is a photon energy, Landauer’s thermal bit its freezing.
2. **Photon and connection.** The gauge potential A_μ is the $U(1)$ connection; the photon is its quantum. One gauge-invariant observable is the holonomy ($\oint A dl$, Aharonov-Bohm, even where $F = 0$) – alongside energy $\hbar\omega$, momentum and polarization, which are not holonomies. The link to “it from connection” is *conceptual* (relational character), not a formal identity with θ .
3. **Counterexample to “mass = information”.** Zero rest mass, yet carrier of essentially all our information transmission.

The neutrino is only the first correction to this: a “damped photon” with a tiny magnitude instead of exactly zero. In the degenerate limit $M = m \cdot \mathbb{1}$ for any U ($\|M - m\mathbb{1}\| \sim 10^{-16}$), the mixing would leave M , the oscillation content sitting in the propagation/holonomy phase – the same situation the photon realises *exactly* and without a degeneracy hypothesis.

.4 What “amount of information” means: packet, count, energy, capacity

Is a photon the smallest packet? Only *per mode*. A single-frequency photon is a plane wave; a localised photon is a wave packet. Crucially: a time-limited packet keeps the *definite* carrier f_0 (the laser colour, the energy jump); the limitation only produces a band $\Delta f \sim 1/\Delta t$ around f_0 (window effect), not a spectrum. The cycle count is $N = f_0 \Delta t$: at fixed time

window $\propto f_0$, but at fixed cycle count the fractional bandwidth $\Delta f/f_0 = 1/N$ is *frequency-independent*. For a photon from an energy jump, the lifetime τ is the window: the natural linewidth $\Delta f = f_0/Q$ with $Q = \omega_0\tau = 2\pi N$ ($\tau = 10$ ns, $f_0 = 5 \cdot 10^{14}$ Hz $\Rightarrow N = 5 \cdot 10^6$, $\Delta f \approx 16$ MHz; numerically `fenster_zyklen_linienbreite.py`). Definite carrier *plus* window gives the band.

Neither cycle count nor window length is intrinsic. The emission process sets both, not “photon-hood”: a monochromatic mode has infinitely many cycles and an infinite window (plane wave); a spontaneously emitted photon inherits f_0 *and* τ from the transition ($N = f_0\tau = Q/2\pi$, the two coupled); a technical packet fixes one quantity, the other follows from $N = f_0\Delta t$. Intrinsic – scale-invariant – is the dimensionless number N ($\Delta f/f = 1/N$); $\Delta t = N/f$ is its dimensional dressing. The span is large: few-cycle pulse $N \approx 2$, narrow atomic line $N \approx 5 \cdot 10^6$ (numerically `quant_bit_modeninfo.py`) – there is no universal cycle count and no universal window length.

Three quantities then separate cleanly (numerically `photon_paket_count_energie.py`):

- **Count:** the quantum number (occupation) is a *count* – one excitation, frequency-independent, relational –, *not* a bit measure. What a quantum carries as *information* is below.
- **Energy** per bit: $E = \hbar\omega = \hbar c/L$ – frequency-dependent (radio $\sim 2 \cdot 10^{-10}$ eV to gamma $\sim 2 \cdot 10^5$ eV).
- **Capacity:** Shannon $C = B \log_2(1 + \text{SNR})$, $B \propto f$ – frequency-dependent ($\sim 2 \cdot 10^5$ to $\sim 2 \cdot 10^{20}$ bit/s).

“Higher frequency $\rightarrow$ more information” is exactly the capacity: more bandwidth, more modes per second. That is a *different* axis from magnitude/phase: *within* a mode each symbol carries amplitude *and* phase (I/Q, QAM), *across* modes the bandwidth counts. FFGFT mirrors both: within the mode the Z_3 -circulant (r, θ) , across modes $|\mathcal{W}|$.

A quantum is a count, not a bit. “One quantum = one bit” is not a general equation – it confuses counting with measuring. A quantum is one *excitation* (a count of 1); its *information content* is $\log_2 M$ bit when it is chosen from M distinguishable modes – *variable*, capped per degree of freedom at $I_1 = \log_2(1/\xi) \approx 12.9$ bit (§5). “One quantum = one bit” holds only for $M = 2$ (the mode fixed to a binary alphabet) – a coding convention, not a law. Count and bit content are different axes: one quantum among 2^{13} modes carries 13 bit, and so do 13 binary quanta – same bit count, different quantum number (numerically `quant_bit_modeninfo.py`). The quantum *counts*; $\log_2 M \leq I_1$ *measures*. This also places §5 cleanly: I_1 is not the content of *one* quantum but the absolute *ceiling* per degree of freedom, which the variable mode content $\log_2 M$ never exceeds.

.5 The absolute amount: anchored in ξ

An earlier reading – information content is representation-relative, there is no absolute bit count – was overstated and unfaithful to FFGFT. The absolute, frequency-independent content is the *dimensionless* quantity, and ξ anchors it. $\xi = 4/30000$ is dimensionless (the absolute parameter); $\hbar, c$ are SI bookkeeping.

ξ caps the winding number at $n_{\max} \sim 1/\xi$ (Doc. 243/009) – without this bound $I = \log_2 |\mathcal{W}|$ diverges. One winding coordinate can take $\sim 1/\xi$ distinguishable values: $I_1 = \log_2(1/\xi) = \log_2(7500) \approx 12.87$ bit. Total $I_{\text{total}} \sim N_{\max} \cdot \log_2(1/\xi)$ (cells $N_{\max} = V/L_0^d$, Doc. 251, times depth). Dimensionless, set by ξ – the energy $\hbar c/L$ is not the amount but its dimensional price, and the capacity saturates at the ξ scale.

Only *weakly* relative are the bit count of a *description* at a chosen resolution and what a lossy projection discards (Doc. 243) – a statement about approximations, not about the information itself. The full information (the geometry, or its lossless Hilbert image, Doc. 230) is absolute.

Absolute and relational are not in conflict. The two statements – the amount is *absolute* (§5), the open datum θ is *relational* (§8) – lie on different axes: “absolute” measures the *amount*, “relational” describes the *nature* of a datum. A holonomy (winding) is the paradigm of *both*: relationally defined (a loop integral, not a local point value) *and* absolutely valued (gauge-invariant, integer). Numerically (absolut_relational.py): a winding $w = 3$ is readable only as a loop integral (the half loop gives 1) and stays unchanged under a single-valued rephasing (gauge-invariant); and the absolute number $\log_2(1/\xi)$ *counts* exactly such relational objects (~ 15000 winding sectors up to $n_{\max} \sim 1/\xi$, $\log_2 \approx 13.9$ bit). The absoluteness sits in the ground (the geometry, ξ), the relationality in the degrees of freedom upon it – the absolute bit count counts relational windings.

.6 Static and dynamic definition of the bit

The definition so far is throughout *static*: the bit as structure – winding $\Delta w = 1$, counting ceiling $\log_2 M \leq I_1$, energy $E_0 = \hbar c/L$ (magnitude side). That says *what* a bit is, not what it *does*. The dynamic definition is not an add-on but the conjugate side – and it itself splits into two, in which energy plays *different* roles.

Kinematic (reversible), the $\tilde{T}$ side: a flip takes $\tilde{T} = \hbar/E_0 = 1/m_0$, the rate is $1/\tilde{T} = E_0/\hbar$. Here $E_0 \cdot \tilde{T} = \hbar$ is nothing but $\tilde{T} \cdot m = 1$ – the time-mass duality, the FFGFT foundation. The quantum speed limit (Margolus-Levitin) $\tau \geq \pi\hbar/2E$ sharpens it: energy *sets the clock* without being dissipated (unitary). The cost is *time*. The capacity axis (bit/s, §4) is exactly this rate.

Thermodynamic (irreversible), the Landauer side: a bit carries entropy $k_B \ln 2$; *erasing* it (forgetting) into a bath at temperature T dissipates $k_B T \ln 2$ of heat. This depends on T , *not* on E_0 , and the heat is *dissipated* – the cost is *heat*, not time. In FFGFT this side is *emergent*: the fundamental T^4 dynamics is unitary-reversible, so the kinematic ($\tilde{T}$) face is primary and the Landauer price arises only through coarse-graining, a bath, and logical irreversibility.

The dimensionless count $\log_2 M$ is primary; the constants dress it (numerically bit_statistisch_dynamisch.py): $\hbar, c$ into energy E_0 and time $\tilde{T}$ (kinematic, reversible: *time* cost, $\tilde{T} \cdot m = 1$), k_B into entropy $k_B \ln 2$ and erase heat $k_B T \ln 2$ (thermodynamic, irreversible: *heat* cost). $\hbar, c, k_B$ are SI bookkeeping. Energy plays two roles: *clock* (kinematic, no loss) versus *dissipated heat* (thermodynamic).

Important: “bookkeeping” does not mean *dispensable*. $k_B, \hbar, c$ are the *necessary* unit bridges. SI separates temperature (K), energy (J), time (s) into their own units, and as soon as one *computes* thermodynamically – and that happens in SI – k_B is the indispensable conversion K→J (otherwise the erase heat would stand in kelvin rather than joule), just as $\hbar, c$ carry the conversion to $E_0 = \hbar c/L$. In natural units ($k_B = \hbar = c = 1$) the costs are pure numbers ($\ln 2$); SI introduces the unit walls, and the bookkeeping is their mortar – ontologically secondary, operationally indispensable (numerically buchhaltung_si_natuerlich.py).

The bit in SI numbers. Anchored at the framework’s characteristic energy $E_0 = \sqrt{m_e m_\mu} = 7.348 \text{ MeV}$ (coupled to the fine-structure constant via $\alpha = \xi E_0^2 / K_{\text{frak}}$; the variant $\sqrt{\alpha/\xi} = 7.398 \text{ MeV}$ is +0.68% higher – up to $7 \cdot 10^{-5}$ the factor $1/\sqrt{K_{\text{frak}}} = 1.0067$, R72). (*Corrected on 30 Sept. 2026: previously “via $\alpha = \xi E_0^2 \dots$ not exact equality” – with the bare E_0 , $\alpha = \xi E_0^2 / K_{\text{frak}}$ holds, and the 0.68% is the factor $K_{\text{frak}}^{-1/2} = 1.0067$.) The generic $E_0 = \hbar c/L$ coincides with*

this E_0 exactly at $L = L_0 = \hbar c/E_0$. Then (numerically `bit_in_si.py`): $E_0 = 1.177 \cdot 10^{-12}$ J, $L_0 = 2.686 \cdot 10^{-14}$ m = 26.85 fm, $m_0 = 1.310 \cdot 10^{-29}$ kg, $\tilde{T} = \hbar/E_0 = 8.958 \cdot 10^{-23}$ s = L_0/c and $1/\tilde{T} = 1.116 \cdot 10^{22}$ s $^{-1}$.

.7 The temperature of the bit: scale, state, bath

The energy E_0 converts directly into a *temperature*. From $E_0 = k_B T_0$,

$$T_0 = \frac{E_0}{k_B} = \frac{1.177 \cdot 10^{-12} \text{ J}}{1.381 \cdot 10^{-23} \text{ J/K}} = 8.53 \cdot 10^{10} \text{ K}$$

(check via 1 eV = 11 604.5 K: $7.348 \cdot 10^6 \text{ eV} \cdot 11 604.5 = 8.53 \cdot 10^{10} \text{ K}$), about 85 billion kelvin. T_0 is the temperature at which the thermal energy $k_B T_0$ reaches the *structure energy* E_0 of the bit.

Does a bit even have a temperature? Strictly: no, not *one*. A single bit in a definite state (0 or 1) has zero entropy and no distribution – temperature is an ensemble/reservoir quantity, not assigned to a single fixed degree of freedom. What one *can* sensibly assign to a bit are *three distinct* temperature-like quantities that must not be confused. First, the *state* or occupation temperature: if the bit has two levels split by E_0 , the occupation ratio $p_1/p_0 = e^{-E_0/k_B T}$ defines an effective temperature. It runs *against* intuition: maximally mixed ($p_0 = p_1 = \frac{1}{2}$, a full one bit of entropy) means $T_{\text{eff}} \rightarrow \infty$, definite ($p \rightarrow 0$ or 1, zero entropy) means $T_{\text{eff}} \rightarrow 0$, population inversion even $T_{\text{eff}} < 0$. The *most informative* bit is thus the infinite-temperature state; cooling drives it to a definite state and *destroys* the information. (A purely logical, degenerate bit – both values same energy – has no occupation temperature at all.) Second, the *scale* $T_0 = E_0/k_B$: not “where the bit sits”, but the characteristic melting/activation point from the energy gap, the natural temperature *unit* of the bit. Third, the *bath* temperature that enters the erase cost – and that belongs to the *environment*, not the bit.

The scale T_0 marks the thermal robustness threshold: above it ($k_B T > E_0$) the thermal energy overwhelms the structure, and the heat washes out the distinction – the winding or mode that the bit *is*: the bit “melts”. Below it ($k_B T < E_0$) it is robust. This is a statement about the *scale*, not about a state.

Which temperature when erasing? The Landauer cost $Q = k_B T \ln 2$ depends on the temperature of the *bath* into which the erased entropy flows as heat – on the choice of environment, not on the bit. There is no universal “correct” value. For a real computer it is the operating temperature, in practice *room temperature* (~ 300 K): $Q = k_B \cdot 300 \cdot \ln 2 = 18$ meV. That is the honest answer for real computing. UIFT instead uses $T_{\text{CMB}} = 2.7$ K – not absolute zero, but the *coldest natural heat sink*; this gives the minimum possible erase heat in the present universe ($Q = 0.16$ meV, nothing colder without active cooling). Legitimate as a cosmological floor, but still just *one* bath choice, not a bit property. T_0 finally is *not* a bath: inserting it gives the limiting case “bath as hot as the bit itself”, $Q = E_0 \ln 2 = 0.69 E_0$ – a reference, not an erasure.

“The” temperature of a bit therefore depends on its *scale*: a generic bit of length L carries energy $E = \hbar c/L$ and hence scale $T = \hbar c/(k_B L)$. Each bit scale has its own; $8.5 \cdot 10^{10}$ K is the one belonging to the characteristic E_0 . A larger L – a “softer” bit – has a lower threshold.

So T_0 stands alongside mass and time as a third unit clothing of the *same* energy. With one bookkeeping constant each:

$$m_0 c^2 = E_0 = k_B T_0 = \hbar / \tilde{T},$$

i.e. $/c^2 \rightarrow \text{mass } m_0 = 1.310 \cdot 10^{-29} \text{ kg}$, $/\hbar \rightarrow \text{time } \tilde{T} = 8.958 \cdot 10^{-23} \text{ s}$ (the dual time from $\tilde{T} \cdot m = 1$), $/k_B \rightarrow \text{temperature } T_0 = 8.53 \cdot 10^{10} \text{ K}$. Beneath all four sits the dimensionless count $\log_2 M$. One energy, four clothings.

The temperature is not merely “also there” but tied directly to the dynamic dual time. The *thermal time* at T_0 is $\hbar/(k_B T_0)$; inserting $T_0 = E_0/k_B$ gives $\hbar/E_0 = \tilde{T}$ – the thermal time at T_0 is the dual time from $\tilde{T} \cdot m = 1$ (numerically identical to 10^{-38} s , `bit_temperatur.py`). This is the KMS / “thermal time” correspondence (Connes–Rovelli): temperature is an inverse (imaginary) time, and it lands without extra assumption exactly on the FFGFT dual time. On the bookkeeping axis k_B is for temperature exactly what $\hbar$ is for time.

Finally, T_0 joins the two “energies” of a bit – what it *is* (E_0) and what its *erasure* costs ($k_B T \ln 2$) – into *one* scale. The Landauer erase heat relative to the self-energy is simply the temperature ratio,

$$\frac{Q}{E_0} = \frac{k_B T \ln 2}{k_B T_0} = \frac{T}{T_0} \ln 2.$$

At room temperature: $\frac{300}{8.53 \cdot 10^{10}} \ln 2 = 2.4 \cdot 10^{-9}$, hence $E_0/Q = (T_0/T)/\ln 2 \approx 4 \cdot 10^8$ – room temperature sits about 8.5 decades below T_0 , and *that* is why the static self-energy ($\sim 7.3 \text{ MeV}$) and the erase heat ($\sim 18 \text{ meV}$) lie a good eight orders of magnitude apart. At $T = T_0$, by contrast, erasure costs $E_0 \ln 2 = 0.69 E_0$: at its own temperature point, *being* and *erasing* are the same order of magnitude. This ratio T/T_0 is the *only* setup-independent statement: a bit has no absolute temperature, only its bath temperature relative to its own scale. The static (E_0, T_0) and thermodynamic ($k_B T \ln 2$) sides are thus joined by the single ratio – no jump, one scale.

Placement: $T_0 \approx 8.5 \cdot 10^{10} \text{ K}$ is the MeV/nuclear scale – the temperature order of nuclear and hadronic physics: stellar cores (solar core $\sim 1.5 \cdot 10^7 \text{ K}$) and the quark-gluon plasma in heavy-ion collisions (QCD transition $\sim 1.7 \cdot 10^{12} \text{ K}$ at 150 MeV). The “hot early universe” reading is deliberately left out – it is a Λ CDM assumption that FFGFT does not share (FFGFT’s CMB structure follows from the T^4 geometry, not from a hot early phase). Honestly: $T_0 = E_0/k_B$ is the *standard* energy-temperature assignment; what is FFGFT-specific is only that it coincides with the dual time via $\tilde{T} \cdot m = 1$ – a clean consistency, not a new derivation. And because the absolute erase temperature is the bath’s, a ξ tied to a particular bath (T_{CMB} or room temperature) would depend on that choice – exactly why the CMB-bound ξ_{UFT} is a contingent, different quantity while FFGFT’s geometric ξ stays bath-*independent* (§9).

E_0 is a rung, not the floor. The scale E_0 is not the bottom end of the framework but a *rung* of a fractal ladder. The geometric floor lies at the sub-Planck length $\xi \ell_P \approx 2.2 \cdot 10^{-39} \text{ m}$ (Doc. 163) – the smallest length below which spacetime carries no well-defined degrees of freedom. Because $\xi \ell_P$ is a factor ξ *below* the Planck length, this floor stands, in the “hard” clothings, a factor $1/\xi = 7500$ *above* Planck:

Length	$\xi \ell_P$	$2.16 \cdot 10^{-39} \text{ m}$
Time/duration	ξt_P	$7.19 \cdot 10^{-48} \text{ s}$
Frequency	$1/(\xi t_P)$	$1.39 \cdot 10^{47} \text{ Hz}$
Energy	E_P/ξ	$9.16 \cdot 10^{22} \text{ GeV}$
Temperature	T_P/ξ	$1.06 \cdot 10^{36} \text{ K}$
Mass	m_P/ξ	$1.63 \cdot 10^{-4} \text{ kg}$

This floor is the *smallest-length* edge, hence in energy/temperature/frequency the *maximum* (UV): “floor” means finest resolution, not smallest energy. The bit rung E_0 (7.348 MeV, 26.85 fm, $T_0 = 8.5 \cdot 10^{10} \text{ K}$) sits about 25 decades *coarser*. Between them the fractal runs inward, factor $1/\xi = 7500$ per main level (Doc. 146), with $\approx 13 \sqrt{2}$ sub-steps each – and $\log_2 7500 \approx 12.87$ is *exactly* the information ceiling I_1 per degree of freedom from §5. The number of fractal

sub-steps per level is the maximum bit count; the capacity of the bit and the fineness of the fractal are the same number. (Note of 30 Sept. 2026: $(\sqrt{2})^{13} = 90.5 \neq 7500$; the factor 7500 requires $\log_{\sqrt{2}} 7500 = 2 \log_2 7500 = 25.7$ steps of $\sqrt{2}$ each, or 12.87 steps of factor 2. Equating the number of $\sqrt{2}$ sub-steps with I_1 therefore does not hold; it holds only for factor-2 steps. The same mix-up appears in Doc. 146.)

Landauer. Erasing one bit is logically irreversible: two states are mapped to one, the entropy drops by $k_B \ln 2$. By the second law this entropy must flow to the environment as heat, $Q \geq k_B T \ln 2$ (Landauer 1961). Here T is the temperature of the *bath* into which the erasure goes – not the bit's.

The problem. And here it jams. The framework now supplies a whole ladder of *definite* temperatures – the rung $T_0 = 8.5 \cdot 10^{10}$ K, the floor $T_{\max} = 1.1 \cdot 10^{36}$ K, each fractal level its own in between, over 25 decades. *None* of them is “the” Landauer temperature. Insert an intrinsic scale into $Q = k_B T \ln 2$ and you get no real erasure but a limiting case or nonsense: with T_0 the reference $Q = 0.69 E_0$, with $T_{\max}$ an absurdly large heat, with T_{CMB} the UIFT number (§9). The real erasure takes the temperature of the *actual environment* (computer: ~ 300 K). That is the problem: despite a ladder of definite intrinsic temperatures, the erase temperature remains the bath's, and the only invariant is the *ratio* $Q/E = (T/T_{\text{rung}}) \ln 2$ – bath relative to the respective rung. A bit has a definite energy and, per rung, a definite scale temperature; a definite *erase price* it acquires only with an environment.

.8 Consequences

The faithful description is quantum, not bit-based. Because the Hilbert space is a bijection (Doc. 230) but the classical bit count a projection: FFGFT's lossless description is the amplitude-plus-phase representation (quantum information); the Shannon bit count is its lossy shadow. An information primitive needs magnitude *and* phase.

The open datum is relational information. The one irreducible open quantity (Doc. 282) is the holonomy phase θ – path-dependent, relational, not a local bit content. The residue the bit projection discards *is* the phase; and the phase *is* where the open physics lives. The fundamental information sits in the connection: the electromagnetic field A_μ is that connection, the photon its quantum – a conceptual example, not one numerically identical to θ . “It from bit” becomes “it from connection”.

.9 Honest boundaries

Structural diagnosis, not a new derivation (P35). Doc. 255/257 already had Landauer/Vopson; new is the magnitude/phase decomposition and its consequences. The *one exact* point: $E_{\text{bit}} = \hbar c/L$ is a photon energy.

Photon is a gauge boson, neutrino a fermion. The photon analogy holds at the level of masslessness/dispersion, not as an identity (spin 1 vs. $\frac{1}{2}$); the link to the connection belongs to the gauge field A_μ (the photon being its quantum), only in the photon sector. The photon's phase (polarization/gauge holonomy) is the same *category* as θ , not the same number. The photon sits outside the lepton mass operator ($m = 0$ reference).

$\log_2(1/\xi) \approx 12.87$ **is not a magic number.** The content is finiteness and ξ anchoring, not the value; the exact form ($1/\xi$ vs. $2/\xi$, direction/flavour factor) depends on the mode counting. Direction: geometry $\rightarrow$ information, not the reverse.

Energy side: ξ -anchored, the value match is empirical. Theoretically ξ -/geometry-fixed ($E_0 = 7.398$ MeV, Doc. 011); for the SI numbers the empirical value $\sqrt{m_e m_\mu} = 7.348$ MeV is used so the comparison with measured quantities fits – from it $m_0, T_0, \tilde{T}, 1/\tilde{T}$ follow exactly ($\hbar, c, k_B$ are mere bookkeeping, no deviation). The only difference is the dimensionless theory-vs-empirics match α/ξ vs. $m_e m_\mu$ (+0.68%), not a conversion.

Bit thermodynamics: computed, over the bit's own scale T_0 . The bit has the full Landauer computation – erase cost $Q = k_B T \ln 2$, relative to the self-energy $Q/E_0 = (T/T_0) \ln 2$ (§7); at $T = T_0$, $Q = E_0 \ln 2 = 0.69 E_0$, the intrinsic quantity is the bit entropy $\ln 2$. No CMB needed. The UIFT parameter $\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2 / (m_e c^2) \approx 3 \cdot 10^{-10}$ (Doc. 013/243) is the same Landauer form with foreign cosmological scales (T_{CMB} instead of T_0 , m_e instead of E_0) – a different quantity, not FFGFT's geometric $\xi = 1.3 \cdot 10^{-4}$. A different scale choice, not a failure.

Neutrino argument: speculative. It still rests on the speculative degeneracy (Doc. 007) – but it no longer carries the thesis; the photon does.

.10 Balance: not whether, but which channel

The magnitude/phase work shifts the information question from “is information fundamental?” to “which channel?”. Vopson/Landauer are the magnitude channel – real, derived from the T^4 geometry, but blind to the phase. The phase channel (mixing, holonomy, θ) is independent, relational information; in signal processing it is the one that carries the structure. The cleanest anchor is the photon: exactly massless, FFGFT's information unit $E_{\text{bit}} = \hbar c / L$ itself, and the quantum of the $U(1)$ connection – conceptually “it from connection”. The *amount* of information is absolute and anchored in ξ ($I_1 = \log_2(1/\xi) \approx 12.87$ bit); energy and capacity are its dimensional dressings. Only the geometry holds both channels – honestly bounded as diagnosis, not a new derivation.